Uniform Boundedness and Orthogonal Polynomial Convergence

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Abstract

This document explores the convergence of a sequence of orthogonal polynomials to the Bessel function J_0 on the semi-infinite interval $[0, \infty)$, utilizing the framework of the Uniform Boundedness Principle (Banach-Steinhaus theorem). The application of this principle is detailed to establish a foundational stability condition for multiplication operators induced by the polynomials within a Banach space setting.

1 Introduction

Consider a sequence of orthogonal polynomials $\{f_k(x)\}$ that pointwise converges to the Bessel function of the first kind, $J_0(x)$, on the semi-infinite interval $[0, \infty)$. The implications of this convergence in terms of boundedness and stability are analyzed via the Uniform Boundedness Principle.

2 Background

2.1 Banach Spaces

A Banach space is a complete normed vector space where every Cauchy sequence converges within the space. Let X denote a Banach space of real-valued functions defined on $[0, \infty)$.

2.2 The Uniform Boundedness Principle (Banach-Steinhaus Theorem)

The Uniform Boundedness Principle is a cornerstone in functional analysis, asserting that for a family of continuous linear operators from a Banach space X to a normed space Y, pointwise boundedness implies uniform boundedness in operator norm.

Theorem 1

(Uniform Boundedness Principle) *Let $\{T_\alpha\}_{\alpha \in A}$ be a family of continuous linear operators from a Banach space X to a normed space Y. If for every $x \in X$, there exists a constant M_x such that $\|T_\alpha(x)\| \leq M_x$ for all α, then there exists a uniform constant M such that $\|T_\alpha\| \leq M$ for all α.*

3 Application to Orthogonal Polynomials

3.1 Multiplication Operators

Define the multiplication operator T_k associated with each polynomial $f_k(x)$ as follows:

$$T_k(g)(x) = f_k(x)\, g(x), \quad \forall g(x) \in X \tag{1}$$

3.2 Pointwise Boundedness

Given the pointwise convergence of $\{f_k(x)\}$ to $J_0(x)$, we assert that for each x, there exists an M_x such that $|f_k(x)| \leq M_x$ for all k, ensuring pointwise boundedness of the sequence of operators $\{T_k\}$.

3.3 Applying the Uniform Boundedness Principle

By the principle, the pointwise boundedness of $\{T_k(g)\}$ for any $g \in X$ implies a uniform bound on the operator norms, $\|T_k\|$, establishing that the sequence $\{f_k(x)\}$ does not lead to unbounded growth in the norm of functions in X when multiplied by $f_k(x)$.

4 Discussion

The uniform boundedness of the multiplication operators $\{T_k\}$ suggests that the sequence of orthogonal polynomials $\{f_k(x)\}$ maintains bounded behavior within the Banach space X. This foundational stability is a crucial step towards further analysis on the uniform convergence of $\{f_k(x)\}$ to $J_0(x)$ and its implications for function approximation and analysis on semi-infinite intervals.

5 Conclusion

We have leveraged the Uniform Boundedness Principle to affirm the stability and boundedness of a sequence of orthogonal polynomials converging to J_0, laying the groundwork for further exploration of their convergence properties and applications in mathematical analysis and functional approximation.